Colin Aysun

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Education

Massachusetts Institute of Technology — Cambridge, MA
BS: Electrical Engineering with Computing — *Expected 2030*

Oviedo High School — Oviedo, FL
GPA: Weighted: 4.849 / Unweighted: 4.0
Valedictorian (1st out of 566)
Clubs/ECs: FTC Robotics Co-Captain, Competitive Programming Club Vice President, ABC Tutoring Co-founder — *2022 – 2026*

Experience

MIT Kavli Institute Research Lab, Researcher (with Prof. Julien De Wit) — *Sep. 2026 - present*

- Researching Machine Learning for Detecting Hidden Exoplanets through Stellar Heartbeats

Perception & Machine Learning

ML-Enhanced Spectral Sensing (Arduino Nano)

- Researched machine learning-enhanced spectral sensing for color detection under varying ambient lighting using cosine similarity, Random Forest classification, scikit-learn library, and SoftMax comparison with a tuned temperature constant
- Performed graphical analysis on data and probability algorithms to compare each color detection method, yielding strong statistical evidence of an 87% improvement in accuracy for the ML model compared to the baseline model.
- Programmed sensor firmware in Arduino and built a Python-based data pipeline with JSON-formatted data handling for training/analysis
- Documented methodology and results in LaTeX; work received Revise & Resubmit from the Journal of High School Science

Computer Vision Pipelines — FTC Robotics

- Built OpenCV-based vision pipelines with contour mapping and tuned PIDF control for real-time object tracking
- Paired odometry pinpoint computer with fiducial marker recognition for localization tasks across multiple robotics projects
- Trained custom TensorFlow Lite object detection models using the FTC ML toolchain, curating and labeling training frame datasets to improve perception accuracy of our field element

Autonomous Systems & Robotics

Autonomous Object Tracker

- Built a vision-guided robot that autonomously navigates to a red Solo cup using a Logitech webcam
- Implemented an RGB-to-YCbCr image processing pipeline with contour detection to isolate the Solo cup as a red “blob”
- Converted real-time pixel measurements into distance and strafe-offset calculations using the focal length of the camera and calibrated pixel width to real-world width ratios

Hanging System — Dual-Linear Actuators

- Designed a bevel gearbox system to power both linear actuators simultaneously with a single motor
- Employed a magnetic limit switch to automatically detect retraction and extension limits without relying on motor encoder counts, ensuring hardware protection
- Refined positioning of motor, shafts, and U-channels in Fusion 360 to make the robot compatible with the rest of the robot and FTC design constraints
- Enabled passive locking to prevent overpowering the motors

In Progress: Drone Tracker — ArduPilot Flight Computer

- Building a drone that autonomously follows a person by integrating different motion planning and object detection techniques, featuring microprocessors, radio telemetry, ArduPilot firmware, circuitry, a GPS module, and ML incorporation.

Software Projects

General Knowledge Trivia App — Google Play Store

- Built responsive UI using scalable SVGs, stacked-column layouts, an embedded scrollbar to prevent screen overflow, and media queries to adapt layout and sizing across device types
- Implemented dynamic quiz generation (20 questions per session from a 3,000+ question bank) using JSON for data filtering, randomization, and score tracking
- Tested and refined gameplay across multiple devices using Flutter web emulators and Android Studio, ensuring consistent cross-platform performance
- Published the final trivia app to the Google Play Store

Maze Racing Game

- Developed a maze racing game in C++ using SFML, implementing graphics rendering, score file I/O, and multiple game-state displays (menu, countdown, gameplay)
- Implemented recursive backtracking for maze generation and configurable random wall regeneration to create deceptive paths and dynamically scale difficulty
- Applied the Separating Axis Theorem (SAT) and dot-product projections for accurate, smooth collision detection
- Published the completed game on itch.io

Technical Skills

Languages: C++ (certified), Java, Python, JavaScript, Dart, HTML, CSS
Frameworks: TensorFlow, Android SDK
Developer Tools: Git, Android Studio, VS Code, Autodesk Fusion 360, PrusaSlicer
Libraries: OpenCV, SMFL, pandas, scikit-learn, NumPy

Other Interests

Varsity Cross Country and Track & Field: 5K: 15:02, Mile 4:13